

INTERNSHIP PROJECT REPORT

AI-Powered Debt Relief & Financial Recovery Platform

An AI-Powered Web Platform for Automated Debt Analysis, Settlement Prediction & Negotiation

Domain	Google Cloud Generative AI
Team ID	
Team Size	5
Submitted By	Kureti Sai Meghana, Ananya Lakshmi Tanneru, Ch Nikhil Kumar, Rehan Shaik, Rukhya Muskan Shaik
Repository	github.com/smd1807/AI_Powered_Debt_Relief_And_Financial_Recovery_Platform

Submitted in partial fulfillment of the requirements for the Internship Program — 2026

1. INTRODUCTION

1.1 Project Overview

The AI Powered Debt Relief & Financial Recovery Platform is an intelligent, AI-driven web application designed to simplify and automate the debt management and financial recovery process for borrowers. The platform allows users to record and manage their loan details, analyze their overall financial health, generate AI-powered negotiation strategies, and receive settlement recommendations, all within a secure and modern digital environment.

The system is built using a Python/JavaScript stack comprising FastAPI for the backend, SQLite with SQLAlchemy ORM for data persistence, JWT-based authentication, React.js (Vite) for the client interface, and the Google Gemini API for generative AI capabilities, backed by a rule-based fallback engine. Together, these technologies provide a responsive, reliable platform with intelligent financial processing and AI-assisted decision support even when external AI access is unavailable.

Core Usage Scenarios

Scenario 1 – AI-Powered Settlement Recommendation: A borrower adds loan details including outstanding amount, interest rate, and due date. The system analyzes the financial profile and generates a settlement prediction along with debt stress analysis and a financial health score.

Scenario 2 – Intelligent Negotiation Letter Generation: A borrower selects a loan and requests a negotiation strategy. The platform uses the Google Gemini AI model (with rule-based fallback) to generate a professional, lender-specific negotiation strategy and settlement letter.

Scenario 3 – Financial Health Tracking: A borrower accesses the dashboard to monitor DTI ratio, EMI ratio, surplus, and financial stress level, while the platform maintains a history of AI-generated negotiations.

1.2 Purpose

The purpose of this project is to address the widespread problem of unmanaged consumer debt by providing an accessible, private, and intelligent alternative to traditional debt settlement channels. Existing options are either costly (professional advisors) or damaging (ignoring lender communication). This platform bridges that gap by combining automated financial analysis with generative AI, enabling financially stressed individuals to understand their debt position clearly and negotiate confidently, without excessive cost.

2. IDEATION PHASE

2.1 Problem Statement

The following problem statement was derived using the Customer Problem Statement framework (I am / I'm trying to / But / Because / Which makes me feel):

I am (Customer)	I'm trying to	But	Because	Which makes me feel
A borrower managing multiple outstanding loans across different lenders.	Understand my overall financial health and find the best settlement strategy for each loan.	I don't know which repayment or settlement approach is realistic, or how to negotiate with lenders effectively.	Financial calculations (DTI, EMI, surplus) are complicated, and I don't have access to personalized financial guidance.	Stressed, uncertain about my financial future, and worried about becoming debt-free.

2.2 Empathy Map Canvas

An empathy map was created to understand target users from four perspectives — Says, Thinks, Does, and Feels — centered on the persona of a Borrower / Debt Manager.

Aspect	Description
Says	"I am overwhelmed with multiple debts." "I don't understand my financial situation clearly." "I need help finding the best settlement option."
Thinks	"How can I reduce my debt burden?" "What is the best way to repay or settle my loans?" "Will I be able to become debt-free?"
Does	Tracks expenses and loan EMIs; searches for debt relief and settlement options; compares loan offers and repayment plans.
Feels	Stressed and anxious about debt; confused by complex financial terms; hopeful when guided with the right strategy.

2.3 Brainstorming

Each team member contributed ideas addressing the core problem, which were grouped and prioritized by feasibility and importance:

Team Member	Idea / Suggestion	Category
Kureti Sai Meghana	Set up the project using React.js, FastAPI, Python, and SQLite.	Set Up
Ananya Lakshmi Tanneru	Create financial health and debt analysis (DTI ratio, surplus, stress level).	AI Analysis
Ch Nikhil Kumar	Build AI-powered settlement prediction engine.	AI Features
Rehan Shaik	Develop secure user authentication and loan management.	Backend
Rukhya Muskan Shaik	Design the dashboard and AI negotiation letter generator UI.	Frontend

The idea that scored highest on both importance and feasibility — an integrated platform combining a financial engine, AI settlement predictor, and AI negotiation letter generator — was selected as the implemented solution.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey traces the borrower's experience across three stages: Discover & Sign Up, Analyze & Plan, and Take Action & Track — from registering an account, through adding loans and reviewing the financial dashboard, to requesting AI negotiation strategies and tracking progress via the AI history log.

3.2 Solution Requirements

FR No.	Requirement	Description
FR-1	Authentication	Secure registration, login, and JWT-based session handling.
FR-2	Financial Profile Management	Maintain monthly income, expenses, and derived metrics per user.
FR-3	Loan Portfolio Management	Add, view, and delete loans (type, amount, interest rate, due date).
FR-4	Financial Health Engine	Compute DTI ratio, EMI ratio, surplus, and stress level live from current loan data.
FR-5	Settlement Prediction	Rule-based settlement amount and priority recommendation per loan.
FR-6	AI Negotiation Strategy & Letters	Gemini-powered negotiation strategy and settlement letter generation, with rule-based fallback.
FR-7	AI Interaction History	Persist and display a log of past AI interactions per user.
FR-8	Dashboard	Central view of loans, financial health, and settlement status.

NFR No.	Requirement	Description
NFR-1	Performance	Dashboard loads in < 2 seconds; Gemini API calls timeout and fall back after 15 seconds.
NFR-2	Security	JWT authentication, bcrypt password hashing; no plaintext credentials stored.
NFR-3	Reliability	Rule-based fallback engine keeps AI features functional if Gemini API is unavailable.
NFR-4	Usability	Mobile-responsive UI with hamburger navigation at the 860px breakpoint.
NFR-5	Scalability	Indexed foreign keys across all 5 tables to keep queries fast as data grows.

3.3 Data Flow Diagram

External Entities: Borrower (User), Google Gemini AI Service.

Processes: Authentication, Loan & Profile Management, Financial Engine, Settlement Predictor, AI Negotiation.

Data Stores: Users, Loans, Financial_Profiles, Settlement_Records, AI_History (SQLite via SQLAlchemy ORM).

Flow: The borrower authenticates → submits loan/financial data through the React client → FastAPI processes and stores data → the settlement/negotiation services call the financial engine and (optionally) the Gemini API → results are persisted and rendered on the Dashboard, Settlement, and Negotiation pages.

3.4 Technology Stack

#	Layer	Technology	Justification
1	Frontend	React.js (Vite), HTML5, CSS3, JavaScript (ES6+)	Fast, responsive, component-based UI.
2	Backend	Python, FastAPI	High-performance framework with automatic validation and docs.

#	Layer	Technology	Justification
3	Database	SQLite	Lightweight, serverless, easy to set up and portable.
4	ORM	SQLAlchemy ORM	Flexible Python database interaction matching the ER diagram.
5	AI / ML Services	Google Gemini API	AI-driven settlement negotiation and letter generation.
6	API Communication	RESTful APIs	Simple, scalable frontend-backend communication.
7	Development Tools	VS Code, Git, GitHub	IDE and version control across the 5-person team.
8	Deployment	Render (Backend), Vercel (Frontend)	Fast, reliable, free-tier hosting.

4. PROJECT DESIGN

4.1 Problem – Solution Fit

Element	Details
Customer Segment(s)	Borrowers with multiple loans; salaried & self-employed individuals; people at risk of default.
Problems / Pains	High interest burden; multiple loan EMIs; lack of clear settlement options; stress and financial anxiety.
Triggers	Unable to pay EMIs; threat of legal action; salary cut/job loss; increasing debt.
Emotions (Before/After)	Before: stress, confusion, fear. After: relief, confidence, financial stability.
Available Alternatives	Traditional settlement agencies (costly); DIY negotiation (time consuming); generic financial advice (not personalized).
The Solution	A self-service AI platform combining a rule-based financial engine and Gemini AI to analyze debt objectively and generate ready-to-send negotiation content.

4.2 Proposed Solution

Objective: Build an AI-powered platform that analyzes financial health, predicts realistic settlement options, and provides intelligent negotiation support to help borrowers achieve debt relief.

Scope: User authentication, loan management, financial health analysis, AI-based settlement prediction, negotiation letter generation, and a dashboard for tracking progress.

4.3 Solution Architecture

The solution follows a layered architecture: Presentation / Client Layer (React.js) → API Gateway (Uvicorn, FastAPI routing, JWT middleware) → Core Logic Services (Auth Service, Financial/Settlement/AI services) with External API Integration (Google Gemini) → Data/Storage Layer (SQLite via SQLAlchemy ORM).

5. PROJECT PLANNING & SCHEDULING

Work was organized into 6 Epics, broken into Sprints and user stories, and executed by the 5-person team.

Sprint	Epic Focus	Team Member(s)
Sprint-1	Epic 1: Application Setup	Kureti Sai Meghana, Rehan Shaik
Sprint-2	Epic 2: AI Integration & Financial Engine	Ananya Lakshmi Tanneru, Ch Nikhil Kumar
Sprint-3	Epic 3: Database Management	Kureti Sai Meghana
Sprint-4	Epic 4: Frontend Integration & UI	Rukhya Muskan Shaik
Sprint-5	Epic 5: Testing & Optimization	Kureti Sai Meghana
Sprint-6	Epic 6: Deployment	Kureti Sai Meghana

Across all six sprints, the team delivered a fully functional, tested, and deployed application. See the detailed sprint backlog in Project Planning (Folder 4) for the complete user-story-level breakdown.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

ID	Scenario	Result
FT-01	User Registration & Login Validation	Pass
FT-02	Loan & Financial Profile Input Validation	Pass
FT-03	Financial Health Calculation (DTI, EMI, surplus, stress level)	Pass
FT-04	Settlement Prediction	Pass
FT-05	AI Negotiation Strategy & Letter Generation (with fallback)	Pass
FT-06	JWT Session Expiry & Redirect	Pass

86 backend tests (pytest) were executed covering authentication, loan management, financial calculations, settlement prediction, and AI negotiation endpoints, including 15 dedicated input-validation tests. All 86 tests pass.

6.2 Performance Considerations

Performance was evaluated through manual testing, automated test coverage, and code-level design decisions rather than a dedicated load-testing tool run. Dashboard and financial calculation endpoints respond quickly since they run entirely server-side. Gemini API calls are capped at a 15-second timeout with automatic rule-based fallback, and database queries are indexed across all foreign key columns to remain fast as data grows. See Performance Testing (Folder 6) for full details.

7. RESULTS

7.1 Demonstration of Proposed Features

Feature	Status	Remarks
User Authentication	Implemented	JWT-based session with automatic expiry handling.
Loan Portfolio Management	Implemented	Full add/view/delete with live recalculation.
Financial Health Dashboard	Implemented	DTI, EMI, surplus, and stress level render correctly.
Settlement Prediction	Implemented	Rule-based recommendation per loan.
AI Negotiation Strategy & Letters	Implemented	Gemini-powered with rule-based fallback.
AI Interaction History	Implemented	Full history log persisted and displayed.
Notifications & Alerts	Pending	Planned for future scope.
Data Export (Reports)	Pending	Planned for future scope.

Overall, 10 of 12 planned features were fully implemented and demonstrated — a delivery rate of approximately 83% against the original scope, with the remaining two items identified as future scope rather than core requirements.

7.2 Code Quality & Reusability

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Modular backend (routers/services) and frontend (api/components/pages).
Readability	5	Descriptive snake_case naming, consistent style throughout.
Reusability	5	Financial engine, settlement, and AI modules are cleanly reusable.
Documentation / Comments	4	Key modules documented; some complex logic could use more inline comments.
Overall Score	19 / 20	Excellent code quality, backed by 86 passing backend tests.

8. ADVANTAGES & DISADVANTAGES

Advantages

Provides borrowers with an automated, objective assessment of their financial and debt-stress position.
Significantly reduces the cost and time barrier of obtaining settlement negotiation guidance.
Rule-based fallback engine keeps AI features working even if the Gemini API is unavailable.
JWT authentication and bcrypt hashing keep sensitive financial data secure.
Modular, reusable backend and frontend architecture makes the platform easy to extend.
86 backend tests provide strong confidence in correctness across all core features.

Disadvantages

Relies on manually entered financial data rather than real-time bank or credit-bureau integration.
Settlement predictions use rule-based thresholds rather than a model trained on historical outcome data.
No automated email/SMS delivery of negotiation letters yet — users must send them manually.
Backend runs on a free-tier host, which may briefly sleep after inactivity.
Formal load testing under concurrent multi-user traffic has not yet been performed.

9. CONCLUSION

The AI Powered Debt Relief & Financial Recovery Platform successfully demonstrates how generative AI and automated financial analytics can be combined to help financially distressed borrowers understand their debt situation and negotiate with creditors confidently and affordably. Across six sprints, the platform was designed, built, tested, and deployed as a full-stack solution comprising JWT-secured authentication, a rule-based financial engine, an AI-powered settlement prediction system, and an AI-driven negotiation letter generator, built on FastAPI, SQLAlchemy, SQLite, React, and the Google Gemini API.

Functional testing confirmed that all core modules — authentication, loan management, financial health calculation, settlement prediction, and AI negotiation — operate reliably, backed by 86 passing automated tests. The application is live and publicly accessible, with the backend deployed on Render and the frontend on Vercel. While certain enhancements — such as real-time bank integration and automated notifications — remain future work, the current solution stands as a stable, secure, and borrower-friendly platform that validates the core value proposition of AI-assisted debt relief.

10. FUTURE SCOPE

Phase	Planned Enhancement	Expected Impact
Phase 2	Email/SMS notifications and alerts for payments and AI updates.	Improves borrower engagement and timeliness.
Phase 3	Exportable financial and settlement reports (PDF/CSV).	Easier record-keeping and sharing with lenders.
Phase 4	Real-time bank statement integration for automatic loan sync.	Improves data accuracy, reduces manual entry.
Phase 5	Native mobile application and predictive credit-score modeling.	Broader reach and more personalized guidance.

11. APPENDIX

Folder Structure

```
backend/  
  app/  
    main.py  
    database.py  
    models.py  
    schemas.py  
    auth.py  
    routers/  
      users.py  
      loans.py  
      settlement.py  
      dashboard.py  
    services/  
      financial_engine.py  
      settlement.py  
      ai_engine.py  
  requirements.txt  
  
frontend/  
  src/  
    api/client.js  
    components/  
    pages/  
  package.json  
  vite.config.js
```

src/api/client.js (excerpt)

```
import axios from "axios";  
const API_BASE_URL = import.meta.env.VITE_API_BASE_URL || "http://localhost:8000";  
const api = axios.create({  
  baseURL: API_BASE_URL,  
  timeout: 20000,  
  headers: { "Content-Type": "application/json" },  
});  
api.interceptors.request.use((config) => {  
  const token = localStorage.getItem("finrelief_token");  
  if (token) config.headers.Authorization = `Bearer ${token}`;  
  return config;  
});  
export const registerUser = (payload) => api.post("/register", payload).then((r) => r.data);  
export const loginUser = (payload) => api.post("/login", payload).then((r) => r.data);  
export const predictSettlement = (loanId) =>  
  api.get("/settlement-predictor", { params: loanId ? { loan_id: loanId } : {} }).then((r) => r.data);
```

Purpose: Centralized API client that attaches the JWT bearer token to every request automatically and matches the full backend endpoint set (register, login, loans, dashboard, financial health, settlement, negotiation).

Dataset Note

This project does not use a pre-existing external training dataset. Settlement prediction uses a transparent, rule-based financial engine (DTI/EMI/surplus thresholds) rather than a model trained on public data. All application data (users, loans, financial profiles, settlements, AI history) is generated live by users and stored in SQLite.

GitHub & Project Demo Link

GitHub Repository: github.com/smd1807/AI_Powered_Debt_Relief_And_Financial_Recovery_Platform

Live Application: ai-powered-debt-relief-and-financia-three.vercel.app

Backend API: ai-powered-debt-relief-and-financial.onrender.com

Project Demo Video: [FILL IN once recorded]

Team Details

Name	Role
Kureti Sai Meghana	Backend Architecture, AI Integration, Deployment, Repository Management
Ananya Lakshmi Tanneru	Financial Engine Logic & Settlement Prediction
Ch Nikhil Kumar	AI Negotiation Strategy & Gemini Integration
Rehan Shaik	Authentication & Security
Rukhya Muskan Shaik	Frontend UI/UX